UNIT-II

MACHINES & MEASURING INSTRUMENTS

Faraday's law and Fleming's right hand rule

Faraday's law states that a current will be induced in a conductor which is exposed to a changing magnetic field. Faraday's law of electromagnetic induction (referred to as Faraday's law) is a basic law of electromagnetism predicting how a magnetic field will interact with an electric circuit to produce an electromotive force (EMF). This phenomenon is known as electromagnetic induction.

Faraday's law of induction explains the working principle of transformers, motors, generators, and inductors. The law is named after Michael Faraday, who performed an experiment with a magnet and a coil. During Faraday's experiment, he discovered how EMF is induced in a coil when the flux passing through the coil changes.

Faraday's First Law

Any change in the magnetic field of a coil of wire will cause an emf to be induced in the coil. This emf induced is called induced emf and if the conductor circuit is closed, the current will also circulate through the circuit and this current is called induced current.

Faraday's Second Law

It states that the magnitude of emf induced in the coil is equal to the rate of change of flux that linkages with the coil. The flux linkage of the coil is the product of the number of turns in the coil and flux associated with the coil.

Fleming's left & right hand rule

Whenever a current carrying conductor comes under a magnetic field, there will be a force acting on the conductor. The direction of this force can be found using Fleming's Left Hand Rule (also known as 'Flemings left-hand rule for motors').Similarly if a conductor is forcefully brought under a magnetic field, there will be an induced current in that conductor. The direction of this force can be found using Fleming's Right Hand Rule.

Principle of a DC Generator

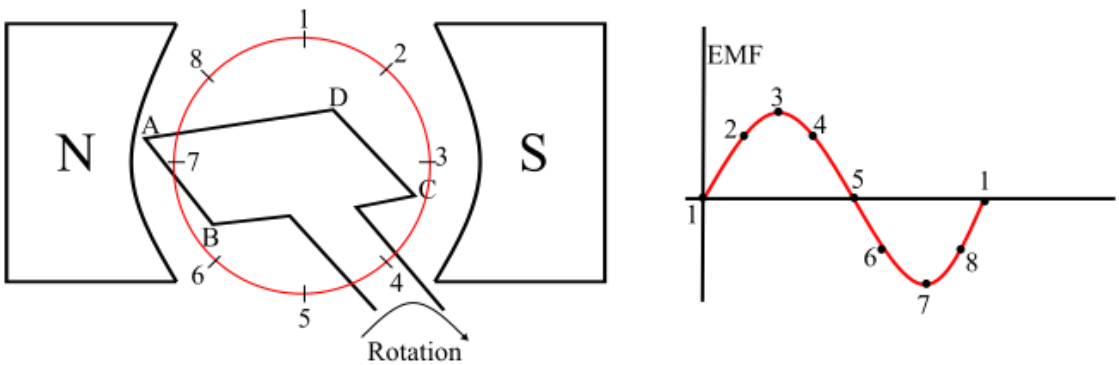
A DC generator is an electromechanical energy conversion device that converts mechanical power into DC electrical power through the process of electromagnetic induction

A DC generator operates on the principle of electromagnetic induction i.e. when the magnetic flux linking a conductor changes, an EMF is induced in the conductor. A DC generator has a field winding and an armature winding.

The EMF induced in the armature winding of a DC generator is alternating one and is converted into direct voltage using a commutator mounted on the shaft of the generator. The armature winding of DC Generator is placed on the rotor whereas the field winding is placed on the stator.

Operation of a DC Generator

Consider a single loop DC generator (as shown in the figure), in this a single turn loop 'ABCD' is rotating clockwise in a uniform magnetic field with a constant speed. When the loop rotates, the magnetic flux linking the coil sides 'AB' and 'CD' changes continuously. This change in flux linkage induces an EMF in coil sides and the induced EMF in one coil side adds the induced EMF in the other.



- When the loop is in position-1, the generated EMF is zero because, the movement of coil sides is parallel to the magnetic flux.
- When the loop is in position-2, the coil sides are moving at an angle to the magnetic flux and hence, a small EMF is generated.
- When the loop is in position-3, the coil sides are moving at right angle to the magnetic flux, therefore the generated EMF is maximum.

- When the loop is in position-4, the coil sides are cutting the magnetic flux at an angle, thus a reduced EMF is generated in the coil sides.
- When the loops is in position-5, no flux linkage with the coil side and are moving parallel to the magnetic flux. Therefore, no EMF is generated in the coil.
- At the position-6, the coil sides move under a pole of opposite polarity and hence the polarity of generated EMF is reversed. The maximum EMF will generate in this direction at position-7 and zero when at position-1. This cycle repeats with revolution of the coil.

It is clear that the generated EMF in the loop is alternating one. It is because any coil side (say AB) has EMF in one direction when under the influence of N-pole and in the other direction when under the influence of S-pole. Hence, when a load is connected across the terminals of the generator, an alternating current will flow through it. Now, by using a commutator, this alternating emf generated in the loop can be converted into direct voltage. We then have a DC generator.

Principle of a DC Motor

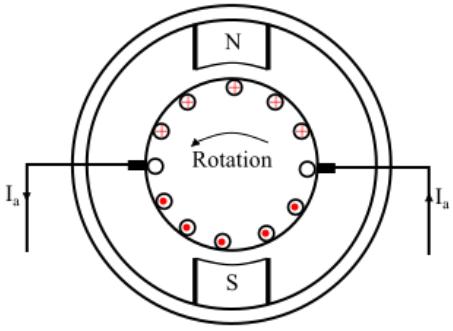
A DC motor is an electromechanical energy conversion device, which converts electrical energy input into the mechanical energy output. The operation of the DC motor is based on the principle that when a current carrying conductor is placed in a magnetic field, a mechanical force acts on the conductor. The magnitude of the force is given by,

$$F=BIl \text{ Newtons}$$

The direction of this is given by the Fleming's left hand rule.

Operation of a DC Motor

Consider a two pole DC motor as shown in the figure. When the DC motor is connected to an external source of DC supply, the field coils are excited developing alternate N and S poles and current flows through the armature windings.



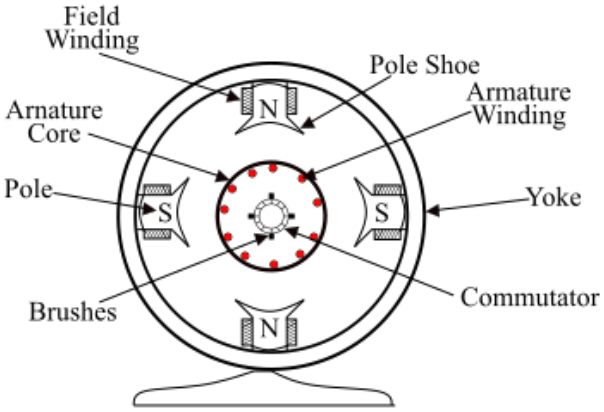
All the armature conductors under N pole carry current in one direction (say into the plane of the paper), whereas all the conductors under S pole carry current in the opposite direction (say out of the plane of the paper). As each conductor carrying a current and is placed in a magnetic field, hence a mechanical force acts on it.

By applying Fleming's left hand rule, it can be seen that the force on each conductor is tending to move the armature in anticlockwise direction. The force on all the conductors add together to exert a torque which make the armature rotating. When the conductor moves from one side of a brush to the other, the current in the conductor is reversed and at the same time it comes under the influence of next pole of opposite polarity. As a result of this, the direction of force on the conductor remains the same. Therefore, the motor being rotating in the same direction.

Construction of a DC Generator (or) Motor

A DC generator consists of six main parts, which are as follows

1. Yoke
2. Magnetic Field System
3. Armature Core
4. Armature Winding
5. Commutator
6. Brushes



Yoke

The outer frame of a DC generator is a hollow cylinder made up of cast steel or rolled steel is known as yoke. The yoke serves following two purposes

- It supports the field pole core and acts as a protecting cover to the machine.
- It provides a path for the magnetic flux produced by the field winding.

Magnetic Field System

The magnetic field system of a DC generator is the stationary part of the machine. It produces the main magnetic flux in the generator. It consists of an even number of pole cores bolted to the yoke and field winding wound around the pole core. The field system of DC generator has salient poles i.e. the poles project inwards and each pole core has a pole shoe having a curved surface. The pole shoe serves two purposes

- It provides support to the field coils.
- It reduces the reluctance of magnetic circuit by increasing the cross-sectional area of it.

The pole cores are made of thin laminations of sheet steel which are insulated from each other to reduce the eddy current loss. The field coils are connected in series with one another such that when the current flows through the coils, alternate north and south poles are produced in the direction of rotation.

Armature Core

The armature core of DC generator is mounted on the shaft and rotates between the field poles. It has slots on its outer surface and the armature conductors are put in these slots. The armature core is a made up of soft iron

laminations which are insulated from each other and tightly clamped together. In small machines, the laminations are keyed directly to the shaft, whereas in large machines, they are mounted on a spider. The laminated armature core is used to reduce the eddy current loss.

Armature Winding

The insulated conductors are put into the slots of the armature core. The conductors are suitably connected. This connected arrangement of conductors is known as armature winding. There are two types of armature windings are used – wave winding and lap winding.

Commutator

A commutator is a mechanical rectifier which converts the alternating emf generated in the armature winding into the direct voltage across the load terminals. The commutator is made of wedge-shaped copper segments insulated from each other and from the shaft by mica sheets. Each segment of commutator is connected to the ends of the armature coils.

Brushes

The brushes are mounted on the commutator and are used to collect the current from the armature winding. The brushes are made of carbon and are supported by a metal box called brush holder. The pressure exerted by the brushes on the commutator is adjusted and maintained at constant value by means of springs. The current flows from the armature winding to the external circuit through the commutator and carbon brushes.

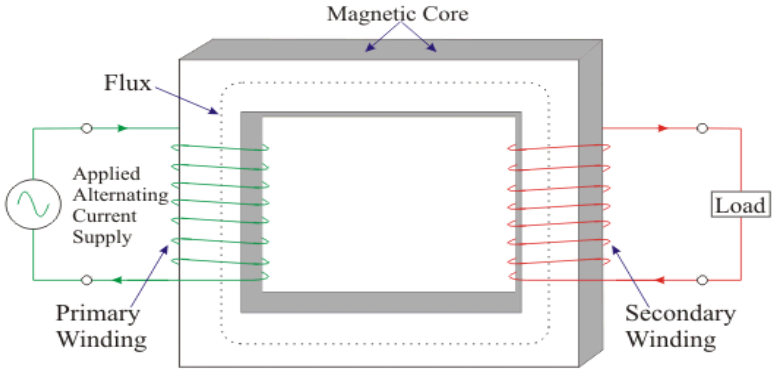
Principle of operation of single phase transformer

Transformer is a static device which transforms electrical power from one circuit to another circuit without change in frequency but change in voltage and current levels according to mutual induction Principle (or) Faraday's law of electromagnetic induction principle.

$$E=M*dI/dt$$

The main principle of operation of a transformer is mutual inductance between two circuits which is linked by a common magnetic flux. A basic transformer consists of two coils that are electrically separate and inductive, but are magnetically

linked through a path of reluctance. The working principle of the transformer can be understood from the figure below.



Single Phase Transformer

When an alternating voltage (V_1) is applied to the primary coil, current flows through the coil which in turn sets up a magnetic field around itself, called self induced emf(E_1) and the flux links with secondary winding it produces e.m.f is called mutual induced e.m.f(E_2) according to Faraday's L aw of electromagnetic induction. A mutual electro-motive force is induced in the transformer from the alternating flux that is set up in the laminated core, due to the coil that is connected to a source of alternating voltage. Most of the alternating flux developed by this coil is linked with the other coil and thus produces the mutual induced electro-motive force.

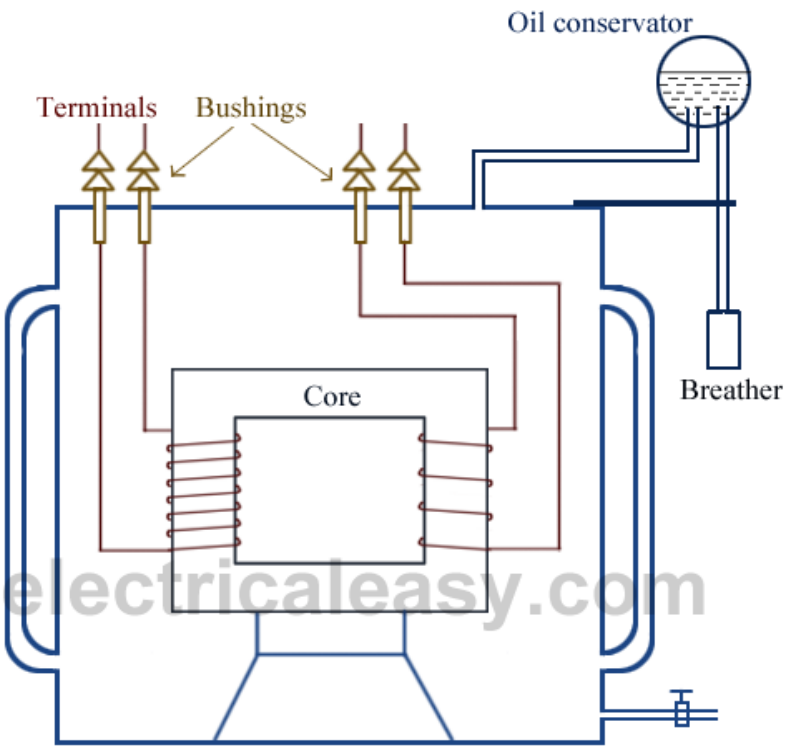
Properties: 1. Transformer is a Constant flux device.

2. Transformer is a Constant Frequency device
3. Transformer is a Constant Power device
4. Transformer is a two port network.
5. Transformer is a phase shifting device
6. Transformer is a –ve Feedback device.

Constructional detail of a single phase Transformer.

These are the basic components of a transformer

1. Laminated core
2. Windings
3. Transformer oil
4. Oil Conservator
5. Breather
6. Cooling tubes
7. Buchholz Relay
8. Explosion vent



Laminated core

The core acts as support to the winding in the transformer. It also provides a low reluctance path to the flow of magnetic flux. It is made of laminated soft iron core in order to reduce eddy current loss and Hysteresis loss.

Windings

Within the input/output supply classification, winding are further categorized:

1. Primary winding - These are the winding to which the input voltage is applied.
2. Secondary winding - These are the winding to which the output voltage is applied.

Within the voltage range classification, winding are further categorized:

1. High voltage winding - It is made of copper conductor. The conductor used will be thinner than that of the low voltage winding.
2. Low voltage winding - It consists of fewer number of turns than the high voltage winding. This is because the current in the low voltage winding is higher than that of high voltage winding.

Oil Conservator

The conservator conserves the transformer oil. It is an airtight, metallic, cylindrical drum that is fitted above the transformer. The conservator tank is vented to the atmosphere at the top, and the normal oil level is approximately in the middle of the conservator to allow the oil to expand and contract as the temperature varies.

Breather

The breather controls the moisture level in the transformer. Moisture can arise when temperature variations cause expansion and contraction of the insulating oil, which then causes the pressure to change inside the conservator

Buchholz Relay

The Buchholz Relay is a protective device container housed over the connecting pipe from the main tank to the conservator tank. It is used to sense the faults occurring inside the transformer. It is a simple relay that is operated by the gases emitted during the decomposition of transformer oil during internal faults. It helps in sensing and protecting the transformer from internal faults.

Cooling tubes

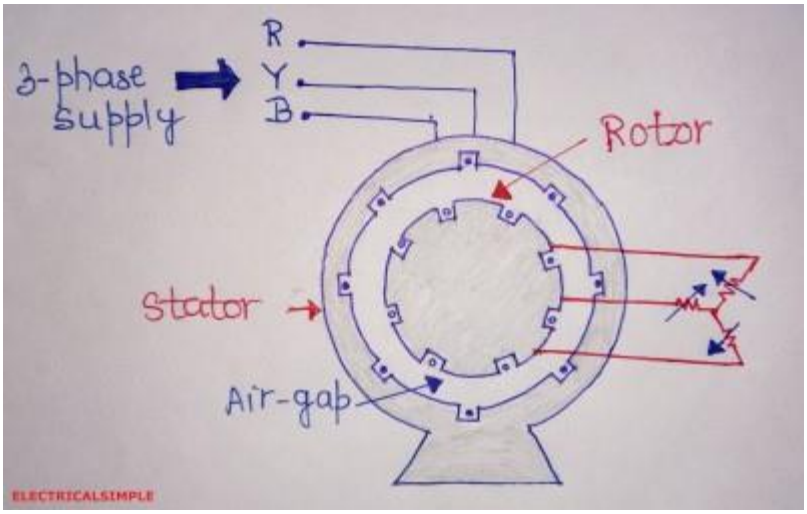
Cooling tubes are used to cool the transformer oil. The transformer oil is circulated through the cooling tubes. The circulation of the oil may either be natural or forced. In natural circulation, when the temperature of the oil raises the hot oil naturally rises to the top and the cold oil sinks downward.

Explosion vent

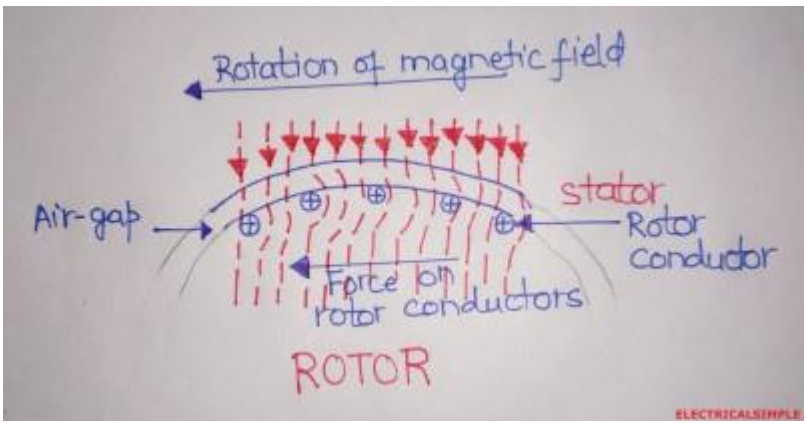
The explosion vent is used to expel boiling oil in the transformer during heavy internal faults in order to avoid the explosion of the transformer.

Principle of operation of 3-Phase Induction Motor

Consider the simplified view of a 3-phase induction motor shown below. The stator hosts a three phase winding distributed symmetrically on its inner periphery. This stator winding is energised from a three phase supply. The rotor also hosts a 3 phase winding on its periphery. But, the rotor winding is not energised from any source and is short- circuited on it self.



1. When the 3 phase stator winding is energised from a 3 phase supply, a rotating magnetic field is produced which rotates around the stator at synchronous speed.



2. The rotating magnetic field cuts the rotor conductors, which as yet, are stationary. Due to this flux cutting, emfs are induced in the rotor conductors. As rotor circuit is short circuited, therefore, currents start flowing in it.

3. Now, as per Lenz's law , “the direction of induced current will be such that it opposes the very cause that produced it “.

4. Here, the cause of emf induction is the relative motion between the rotating field and the stationary rotor conductors. Hence, to reduce this relative motion, the rotor starts rotating in the same direction as that of the stator field and tries to catch it but, can never catch it due to friction and windage and therefore emf induction continues and motor keeps rotating.

Principle of operation Alternators (or) **synchronous *generator***

A *synchronous generator* is a synchronous machine which converts mechanical power into AC electric power through the process of electromagnetic induction. Synchronous generators are also referred to as alternators or AC generators. The term "alternator" is used since it produces AC power. It is called synchronous generator because it must be driven at synchronous speed to produce AC power of the desired frequency. A synchronous generator can be either *single-phase or poly-phase (generally 3phase).*

An alternator or synchronous generator works on the principle of electromagnetic induction, i.e., when the flux linking a conductor changes, an EMF is induced in the conductor. When the armature winding of alternator subjected to the rotating magnetic field, the voltage will be generated in the armature winding.

When the rotor field winding of the alternator is energised from the DC exciter, the alternate N and S poles are developed on the rotor. When the rotor is rotated in the anticlockwise direction by a prime mover, the armature conductors placed on the stator are cut by the magnetic field of the rotor poles. As a result, the EMF is induced in the armature conductors due to electromagnetic induction. This induced EMF is alternating one because the N and S poles of the rotor pass the armature conductors alternatively.

The direction of the generated EMF can be determined by the Fleming's right rule and the frequency of it is given by,

$$f=N_sP/120$$

The magnitude of the generated voltage depends upon the speed of rotation of the rotor and the DC field excitation current. For the balanced condition, the generated voltage in each phase of the winding is the same but differ in phase by 120° electrical.

Construction of Synchronous Generator or Alternator

As alternator consists of two main parts viz.

- **Stator** – The stator is the stationary part of the alternator. It carries the armature winding in which the voltage is generated. The output of the alternator is taken form the stator.
- **Rotor** – The rotor is the rotating part of the alternator. The rotor produces the main field flux.

Stator

The stator of the alternator includes several parts, viz. the frame, stator core, stator or armature windings, and cooling arrangement.

- The stator frame may be made up of cast iron for small-size machines and of welded steel for large-size machines.
- The stator core is assembled with high-grade silicon content steel laminations. These silicon steel laminations reduce the hysteresis and eddy-current losses in the stator core.
- The slots are cut on the inner periphery of the stator core. A 3-phase armature winding is put in these slots.
- The armature winding of the alternator is star connected. The winding of each phase is distributed over several slots. When current flows through the distributed armature winding, it produces an essential sinusoidal space distribution of EMF.

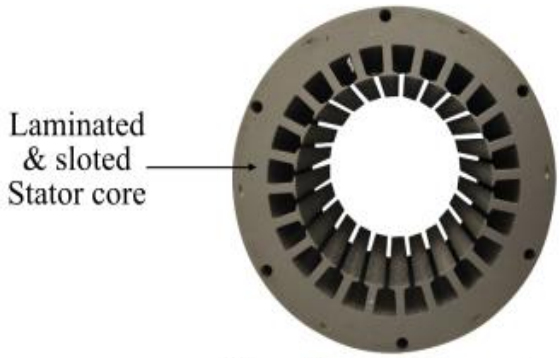


Fig. - Stator of Alternator

Rotor

The rotor of the alternator carries the field winding which is supplied with direct current through two slip rings by a separate DC source (also called exciter). The exciter is generally a small DC shunt generator mounted on the shaft of the alternator. For the alternator, there are two types of rotor constructions are used viz. (i)salient-pole type

*(ii)*cylindrical rotor type.

Salient Pole Rotor

The term *salient means projecting*. Hence, a *salient pole rotor* consists of poles projecting out from the surface of the rotor core. This whole arrangement is fixed to the shaft of the alternator as shown in the figure. The individual field pole windings are connected in series such that when the field winding is energised by the DC exciter, the adjacent poles have opposite polarities.

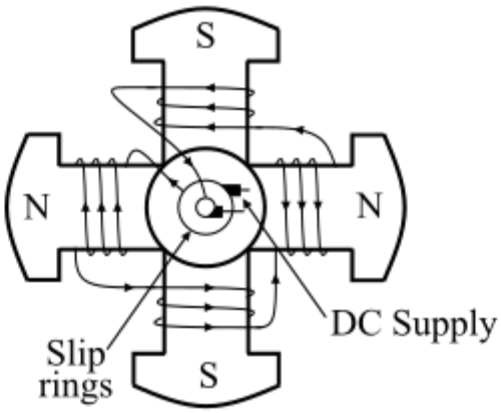


Fig. - Salient Pole Rotor

The salient pole type rotor is used in the low and medium speed (from 120 to 400 RPM) alternators such as those driven by the diesel engines or water turbines because of the following reasons −

- The construction of salient pole type rotor cannot be made strong enough to withstand the mechanical stresses to which they may be subjected at higher speed.
- If the salient field pole type rotor is driven at high speed, then it would cause windage loss and would tend to produce noise.

Low speed rotors of the alternators possess a large diameter to provide the necessary space for the poles. As a result, the salient pole type rotors have large diameter and short axial length.

Cylindrical Rotor

The cylindrical rotors are made from solid forgings of high-grade nickel-chrome-molybdenum steel.

- The construction of the cylindrical rotor is such that there are no-physical poles to be seen as in the salient pole rotor.
- In about two-third of the outer periphery of the cylindrical rotor, slots are cut at regular intervals and parallel to the rotor shaft.
- The field windings are placed in these slots and are excited by DC supply. The field winding is of *distributed type*.
- The unslotted portion of the rotor forms the pole faces.
- It is clear from the figure of the cylindrical rotor that the poles formed are non-salient, i.e., they do not project out from the rotor surface.

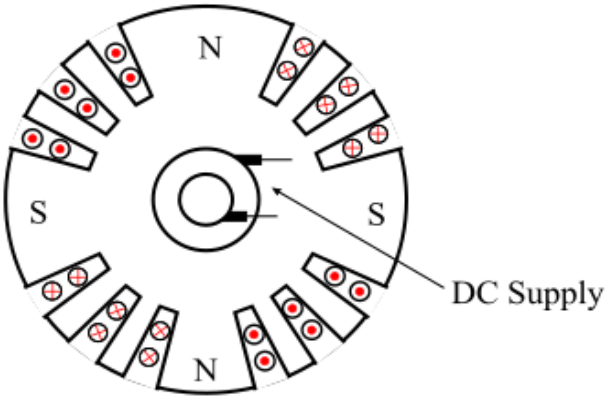


Fig. - Cylindrical Rotor

The cylindrical type rotor construction is used in the high-speed (1500 to 3000 RPM) alternators such as those driven by steam turbines because of the following reasons −

- The cylindrical type rotor construction provides a greater mechanical strength and permits more accurate dynamic balancing.
- It gives noiseless operation at high speeds because of the uniform air gap.
- The flux distribution around the periphery of the rotor is nearly a sine wave and hence a better EMF waveform is obtained.

A cylindrical rotor alternator has a comparatively small diameter and long axial length. The cylindrical rotor alternators are called turbo-alternators or turbo-generators. The alternator with cylindrical rotor has always horizontal configuration installation.

Construction of Three-Phase Induction Motor

A three-phase induction motor consists of two main parts namely,

- Stator
- Rotor

There is a small air gap between the stator and rotor which ranges from 0.4 mm to 4 mm depending on the power rating of the motor.

Stator

The stator of a three-phase induction motor is a stationary part, and it consists of a cylindrical-shaped frame made up of fabricated steel. This steel frame encloses a hollow cylindrical core made up of thin laminations of silicon steel. On the inner periphery of the core, a number of evenly spaced slots are provided to place the stator winding. The silicon-steel laminations are used to reduce the hysteresis and eddy current losses.

Figure 1 - Three-Phase IM Stator

Three windings are placed in the stator slots and are suitably connected to form a balanced three-phase delta or star connected circuit. As per the requirement of motor speed, these three-phase windings are wound for a definite number of poles. Where, greater is the number of poles, lesser is the speed of the induction motor and *vice-versa*.

When we fed the three-phase stator winding from a balanced three-phase supply, a rotating magnetic field of constant magnitude is produced. This rotating magnetic induces EMF in the rotor circuit by electromagnetic induction

Rotor

The rotor is a rotating or moving part of the three-phase induction motor. It consists of a rotor core made up of thin laminations of high grade silicon steel to reduce the hysteresis and eddy-current losses. The rotor core is a hollow cylinder, mounted on a shaft. On outer periphery of the rotor core, slots are provided to place the rotor winding.

Based on the construction, the rotor of a three-phase induction motor can be of the following two types −

- Squirrel-cage rotor
- Wound rotor

Squirrel Cage Rotor

The squirrel-cage rotor consists of a laminated cylindrical core having parallel slots on its outer periphery. In case of squirrel-cage rotor, the rotor winding is made up of metal (copper or aluminium) bars. These metal bars are placed in the rotor slots and are short-circuited at each end by metal rings called end-rings as shown in Figure

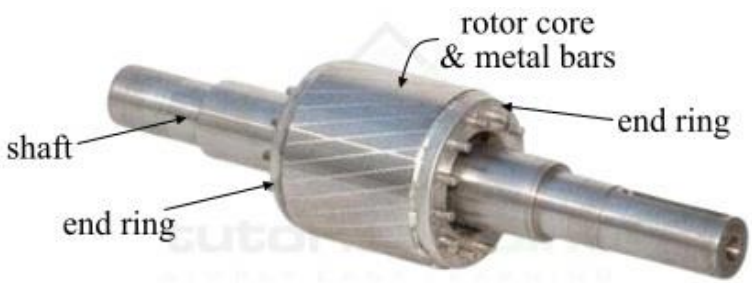


Figure 2 - Squirrel Cage Rotor

The three-phase induction motors that employ squirrel-cage rotor are known as squirrel-cage induction motors. Almost 70% to 80% three-induction motors used in industrial applications are squirrel-cage induction motors because of their simple and robust construction which enable them to operate in most adverse circumstances. Although, the induction motors that use squirrel-cage rotor have a low starting torque.

Wound Rotor

The wound rotor consists of a laminated cylindrical core made up of silicon steel. It carries a 3-phase rotor winding similar to the stator winding as shown in Figure

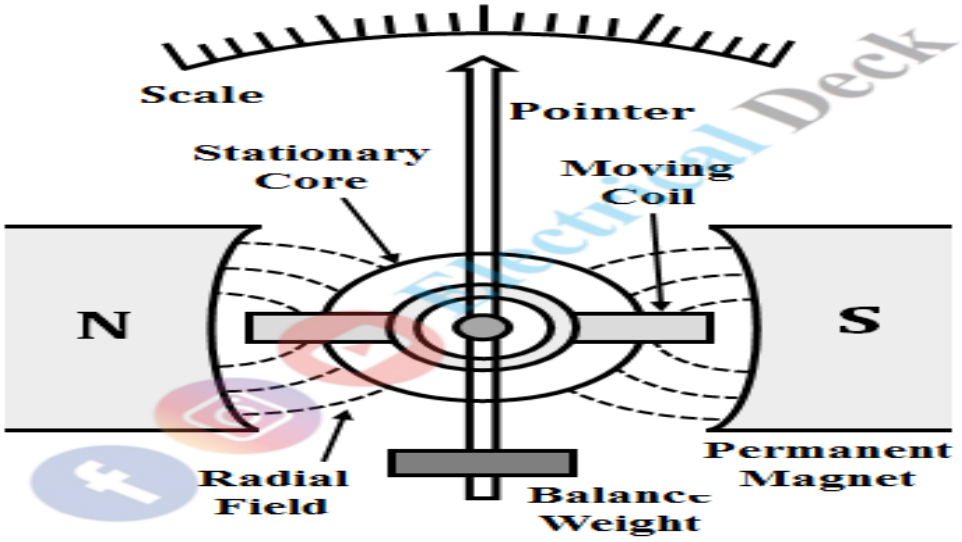
Figure 3 - Wound Rotor

The rotor winding of the wound rotor is uniformly distributed in the slots and is connected in star fashion. The open ends of the star-connected rotor winding are brought out and connected to three slip rings mounted in the rotor shaft. A carbon brush is resting on each slip ring, and through these brushes, external resistances can be added to the rotor circuit.

At starting, suitable values of external resistances are added into phases of the rotor winding to obtain a high starting torque. These external resistances are gradually removed from the circuit as the motor runs up to speed. The use of external resistances considerably reduces the starting current and increases the starting torque of the motor. Once the motor attains normal speed, the three carbon brushes are short-circuited so that the wound motor runs like a squirrel cage induction motor.

Working principle of Permanent Magnet Moving Coil or PMMC Instrument

Whenever a current caring conductor is located within a magnetic field, then it experiences a force that is perpendicular to the current & the field. Based on the rule of "Fleming left hand", if the thumbnail of the left hand, middle, and forefinger are at 90 degrees with each other. After that the magnetic field will be in the forefinger, the flow of current will be across the middle finger and finally, the force will be through the thumb finger. Once the current flow within the coil on the aluminium former, the magnetic field can be generated in the tesla coil in proportion to the current flow.



Construction of the PMCC

The construction of the PMCC instrument can be done using several parts where the permanent magnet and moving coils are essential parts. Each part of this instrument is discussed below.

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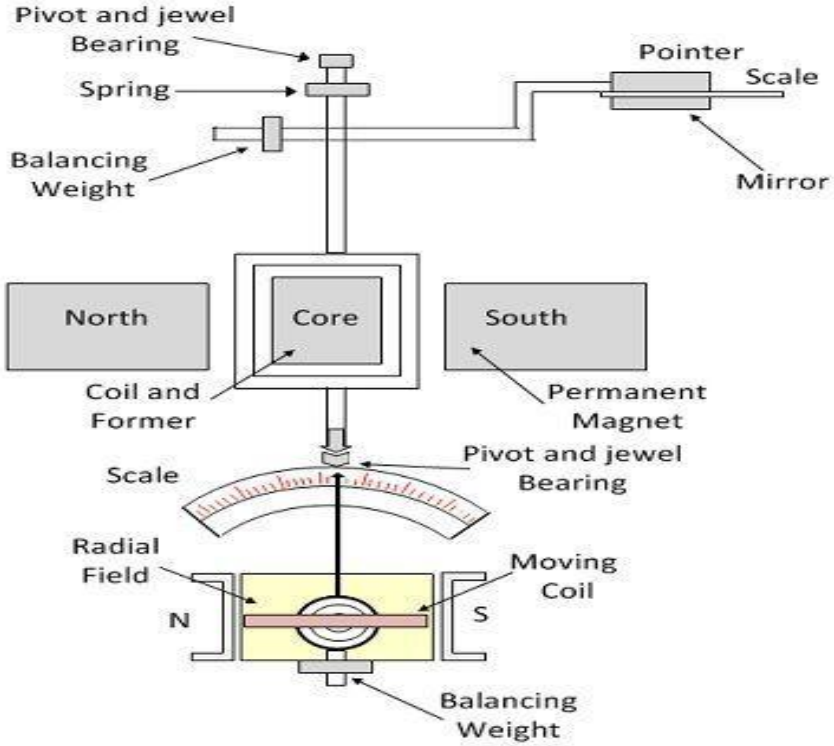


Permanent Magnet Moving Coil Instrument

1. **Moving Coil**: The design of this coil can be done by wounding copper coils to a rectangular block among the magnetic poles. It is made with Aluminium and the rectangular block can be called Aluminium former rotated into the jewel bearing. The aluminium former is a non-metallic former used to measure current, while the metallic former including high electromagnetic damping is used to calculate the voltage.

2. **Magnet system**: Two permanent magnets of high-intensity magnetic field mostly soft iron cylinder used.

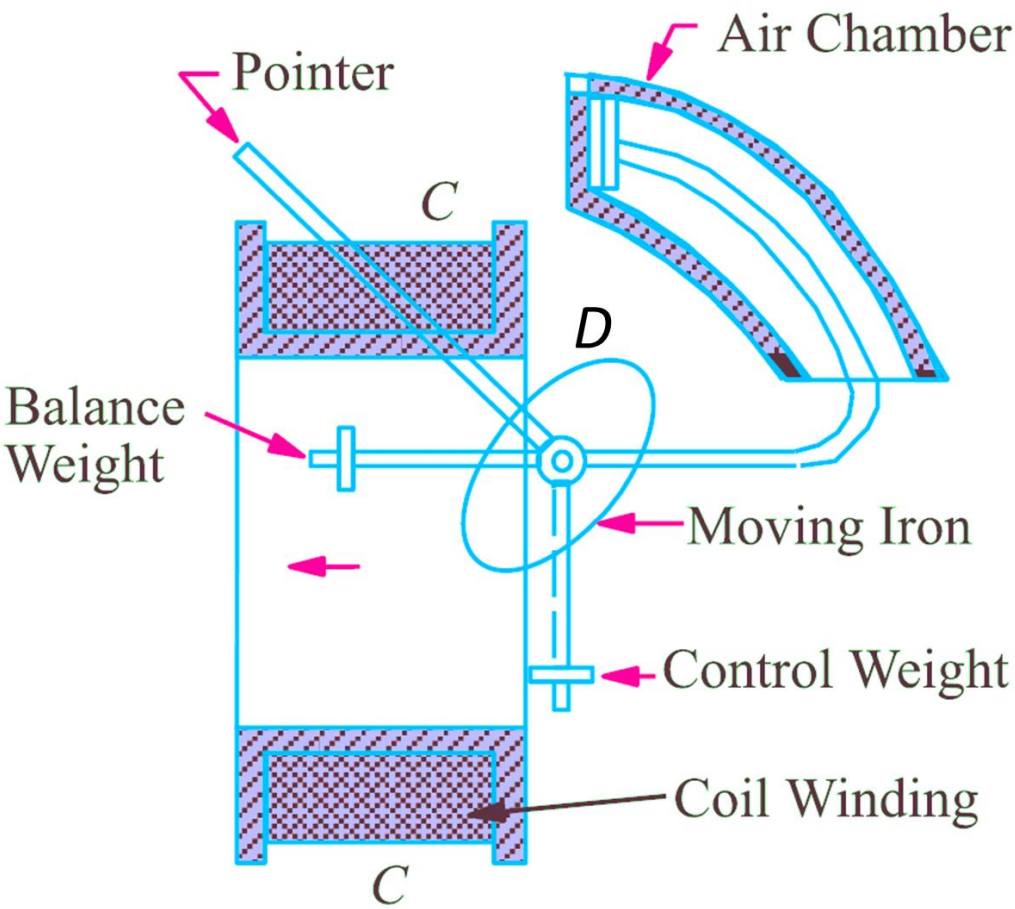
3. **Control spring**: torque can be controlled due to the springs which are fabricated with phosphorous bronze. These springs are arranged among the two jewel bearings. These springs are responsible for producing controlling torque.

4. **Damping torque**: Damping torque can be generated within the PMMC instrument using the aluminium core's movement within the magnetic field. The aluminium frame is responsible for producing damping torque.

5. **Pointer and Scale arrangement**: connection of the pointer can be done through the moving coil. It notices the moving coil's deflection. The magnitude of their derivation can be displayed on the scale. The pointer is designed with lightweight material.

Working principle of moving iron instrument

Moving Iron Instruments are the most common type of ammeter and voltmeter used at power frequencies in laboratories. These instruments are very accurate, cheap and rugged as compared to other AC instruments. In Moving Iron Instruments, a plate or van of soft iron or of high permeability steel forms the moving element of the system. The iron van is so situated that it can move in the magnetic field produced by a stationary coil. Figure below shows a simple moving iron instrument.



The stationary coil is excited by the current or voltage under measurement. When the coil is excited, it becomes an electromagnet and the iron van moves in direction of offering low reluctance path. Thus the force of attraction is always produced in a direction to increase the inductance of coil. Mind that as the van follows the low reluctance path, the net flux in air gap will increase which means increased flux linkage of coil and hence inductance of coil will increase. It shall also be noticed that, the inductance of coil is variable and depends on the position of iron van.